

Total No. of Questions : 6]

SEAT No. :

P210

APR -17/TE/Insem. - 46

[Total No. of Pages : 2

T.E. (Information Technology)

OPERATING SYSTEM

(2012 Course) (Semester - II) (314451)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Neat diagram must be drawn whenever necessary.
- 3) Figures to the right indicate the full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain the differences between multiprogramming, multitasking and multiuser OSs. [6]
- b) Explain operating system functions. [4]

OR

- Q2)** a) Explain the concept of virtual machine with its implementations and benefits.[6]
- b) What are the benefits of resource abstraction. [4]

- Q3)** a) Consider the following set of processes with length of CPU burst time given in milli seconds. [8]

Process	Arrival Time	Burst Time
P1	0	9
P2	1	1
P3	2	7
P4	3	1
P5	4	6

Apply SJF (Preemptive), RR(TQ=1 ms).Calculate average waiting time and average turn around time.

- b) Describe exec family of system call. [2]

OR

P.T.O.

- Q4) a)** Consider the following set of processes with length of CPU burst time given in milliseconds.

Process	Arrival Time	Burst Time
P1	0	9
P2	2	20
P3	4	2
P4	7	14
P5	8	04

Apply FCFS, SJF (Preemptive). Calculate average waiting time and average turn around time. Draw Gantt chart. **[8]**

- b) Draw process state transition diagram. **[2]**

- Q5) a)** Consider following snapshot of system.

Process	Allocation			Max			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			
P3	3	0	2	9	0	2			
P4	2	1	1	2	2	2			
P5	0	0	2	4	3	3			

Answer the following questions using bankers algorithm.

- i) What is the content of matrix need? **[6]**
 ii) Is the system in safe state? What is safe sequence. **[6]**
 b) What are the four conditions for deadlock to occur? **[4]**

OR

- Q6) a)** Write a pseudo code for Reader-writers problem with readers priority. **[5]**

- b) Enlist different IPC mechanisms differentiate between named pipe and unnamed pipe. **[5]**

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